

Multi-Period Average Backtesting Probability of Default Models

Sensitivity of the Z-Score Test to Serial Correlation in Default Rates

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Internal Ratings-Based Modeling Process

- The development of an Internal Ratings-Based (IRB) model consists of two main areas: the development of the risk differentiation function and risk quantification.
- Development of the risk differentiation function is the first step in the overall model development process. The main goal of this step is to ensure the model can adequately rank risk, providing meaningful differentiation between low- and high-risk obligors or facilities.
- After developing the risk differentiation function, practitioners perform the risk quantification process, which essentially involves assigning an appropriate risk level to the output of the risk differentiation function.
- Risk quantification typically consists of specific steps such as: assignment of the base or best estimate, quantification of appropriate adjustments (AA), quantification of the margin of conservatism (MoC), and quantification of any other additional conservatism, for instance that required by certain local regulators. The first two steps are typically referred to as calibration.
- These steps apply to all three risk parameters: Probability of Default (PD), Loss Given Default (LGD), and Exposure at Default (EAD).

Probability of Default Backtesting

- As risk quantification is a regulatory requirement, practitioners pay significant attention to complying with these requirements.
- In addition to adhering to the prescribed risk quantification process, practitioners are required to assess the appropriateness of model estimates. This is typically referred to as backtesting.
- The primary purpose of backtesting in this context is to ensure that model estimates accurately predict the occurrence of defaults, i.e., that PD estimates provide reliable forecasts of default rates.
- In practice, this is one of the most comprehensive validation exercises, typically investigating model estimates across different portfolio subsegments and time dimensions. One commonly applied approach is to backtest estimates over a recent period of three to five years.

Probability of Default Backtesting cont.

- In this exercise, practitioners typically face challenges in formulating the statistical approach. Currently, there is no standardized or recommended approach to multi-period backtesting in credit risk modeling practice, despite its relevance not only for IRB models but also for IFRS9 models.
- One possible approach to multi-period average testing is the use of the z-score. However, to ensure the test is applied correctly, the testing framework should be aligned with the design of the model estimates used in the analysis. Details on the proper implementation of the z-score test for this purpose can be found [here](#) and [here](#).
- This presentation investigates the sensitivity of the z-score test to serial correlation in default rates. Specifically, it examines the size of the Type I error under different dependency structures and significance levels.

The Z-Score Test

The test statistic of the z-score test is given as follows:

$$z_{score} = \frac{ODR - PD}{\sqrt{\frac{PD(1-PD)}{N}}}$$

where:

- ODR is the observed default rate;
- PD is the calibrated PD;
- N is the number of exposures.

Under the assumption that the z_{score} test statistic follows a standard normal distribution, a p-value is calculated accordingly.

The Z-Score Test cont.

- When applying this test for the purpose of multi-period average testing, it is necessary to ensure that the inputs (average ODR and sample size) are aligned with the PD estimate.
- In simple terms, the average ODR should be calculated in the same way as the PD estimate, with the arithmetic average of the reference-date ODRs being the most common approach.
- Additionally, the effective sample size should be aligned with the variance of the average estimator.
- Under the assumption of independence of ODRs over time, it is straightforward to show that the effective sample size is given by $N_e = \frac{T^2}{\sum_{i=1}^T \frac{1}{n_i}}$, where T is the number of reference periods used in the test and n_i is the number of exposures (sample size) at reference date i .
- The assumption of independence will be used in this presentation to demonstrate the impact of serial correlation in default rates on the validity of the z-score test. However, practitioners should be aware that adjustments to the z-score test can be made to properly account for serial correlation in default rates.

Simulation Study

The purpose of this simulation is to examine the sensitivity of Type I error to serial correlation in default rates for different significance levels. Therefore, the simulation relies on the following assumptions:

- 1 the true probability of default (LRADR) is 5%;
- 2 default rates exhibit a first-order autocorrelation structure, with coefficients ranging from 0 to 0.90 in increments of 0.10;
- 3 significance levels are set to 0.01, 0.05, 0.10, 0.15, 0.20, and 0.30.

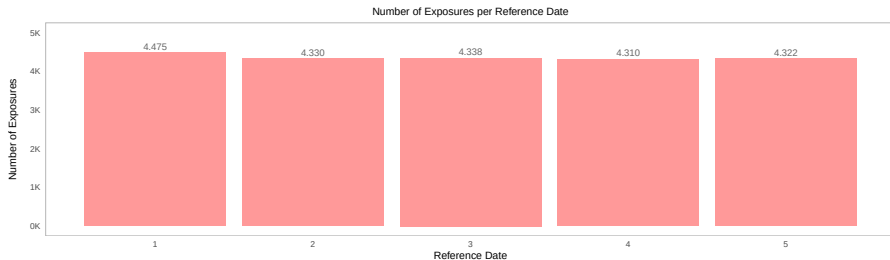
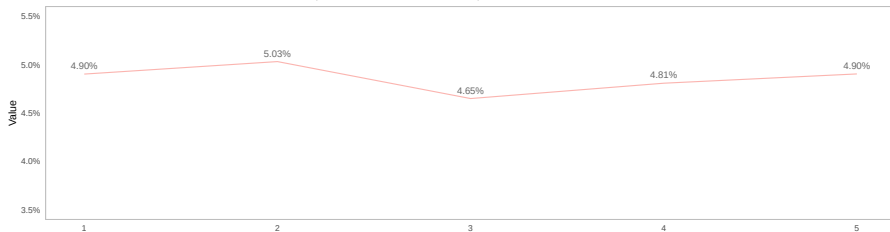
Under these assumptions, and for the numbers of exposures per reference date shown on the next slide, the simulation design consists of the following steps:

- 1 for a selected autocorrelation coefficient, simulate the number of defaults using a Gaussian copula;
- 2 given the number of defaults from step 1, calculate the simulated observed default rate;
- 3 run a z-score test, where the average ODR is defined as the arithmetic mean of the simulated default rates, and the effective sample size is calculated under the assumption of independence;
- 4 repeat steps 1 to 3 10,000 times and store the resulting p-values;
- 5 calculate Type I error as the proportion of simulations where the p-value is less than the selected significance level;
- 6 repeat steps 1 to 5 for all combinations of autocorrelation coefficients and significance levels.

As the above simulation uses a continuous distribution to generate discrete, correlated numbers of defaults, the dependency structure may deviate from the assumed one. Therefore, practitioners are encouraged to improve the above steps to better reflect the underlying assumptions.

Simulation Results

Example of the Observed Default Rates per Reference Date for true PD of 5%



Simulation Results cont.

Type I Error by Correlation and Significance Level

